NORMATIVE

T = I  det = +1  (x, y) → (x, y)

"This is the standard."

RFC 2119 MUST / SHALL

# DNS Tool

Communication Standards — Clarity & Vision Quality Gate

**Carey James Balboa**

Independent DNS Security Researcher

**ORCID** 0009-0000-5237-9065

**DOI** 10.5281/zenodo.18854899

**PROJECT** dnstool.it-help.tech **SOURCE** github.com/IT-Help-San-Diego/dns-tool-web

**VERSION** 26.38.35 · **LICENSE** BUSL-1.1

Companion to Founder's Manifesto and Philosophical Foundations

## The Owl Semaphore System

The Owl of Athena encodes document classification as semantic state through orientation and color. Each designation is a distance-preserving isometry drawn from the orthogonal group O(2). The system uses three transforms to classify every publication in the DNS Tool corpus.

**NORMATIVE**

$T = I \quad \det = +1$

$(x, y) \to (x, y)$

"This is the standard."

**RFC 2119 MUST / SHALL**

**NON-NORMATIVE**

$T = \sigma_v \quad \det = -1$

$(x, y) \to (-x, y)$

"This reflects the standard."

**Informative / Advisory (NOTE)**

**CRITICAL**

$T = C_2 \quad \det = +1$

$(x, y) \to (-x, -y)$

"This inverts the standard."

**RFC 2119 MUST NOT / SHALL NOT**

**Redundant encoding:** Color, orientation, and context label provide three independent channels. The semaphore remains distinguishable at small render sizes and in degraded single-channel contexts. Each transform is a distance-preserving isometry in O(2). **Standards mapping:** Normative = RFC 2119 MUST/SHALL. Non-normative = Informative/NOTE. Critical = RFC 2119 MUST NOT/SHALL NOT; CVE/Security advisory.

## 1. Dual-Gate Requirement

Every piece of user-facing text must pass **both** gates simultaneously. Passing one gate but failing the other is a failure.

### Gate 1: Clarity

The words chosen must be clear, communicative, and logical to the reader. Text should educate, not obscure.

| REQUIREMENT | LEVEL | MEASUREMENT |
|---|---|---|
| Jargon with first use | MUST | Every acronym expanded on first use per page, or linked |
| Active voice | SHOULD | Prefer "The SPF record authorizes..." over passive |
| Sentence length | SHOULD | Body copy sentences should not exceed 35 words |

| | | |
|---|---|---|
| One idea per paragraph | SHOULD | Each paragraph should advance one concept |
| Action-oriented findings | MUST | State what was observed, why it matters, what to do |
| RFC citation | MUST | Claims about protocol behavior must cite specific RFC section |
| Transformative language | SHOULD | Prefer language that teaches rather than restates |

### Gate 2: Vision

The visual presentation must not cause strain or impede comprehension. The reader's eyes matter.

| REQUIREMENT | LEVEL | MEASUREMENT |
|---|---|---|
| Minimum font size | MUST | No rendered text below 0.75rem (12px) |
| Body text minimum | SHOULD | Primary reading text at least 0.875rem (14px) |
| Contrast ratio (normal text) | MUST | WCAG 2.1 AA: minimum 4.5:1 against background |
| Contrast ratio (large text) | MUST | WCAG 2.1 AA: minimum 3:1 for 18px+ or 14px+ bold |
| Text overflow | MUST | No text beyond container boundary |
| Line length | SHOULD | Body text ≤ 80 characters per line |
| Line height | SHOULD | Body text at least 1.5 line-height |
| Scotopic mode | MUST | Minimum 4.5:1 contrast against covert background (#0a0c10) |
| Focus indicators | MUST | Visible focus indicators on all interactive elements |

## 2. Audience Calibration

DNS Tool serves three primary audiences with different reading contexts:

**The Engineer**

**Context:** Analyzing a domain at a terminal, possibly at 2 AM, under incident pressure.
**Needs:** Precision, RFC references, verification commands, machine-parseable output.
**Vision concern:** Extended reading sessions on dark backgrounds. Font size and contrast directly affect error rate.

**The Executive**

**Context:** Reading a brief before a board meeting or vendor assessment.
**Needs:** Risk posture in business terms, remediation cost/complexity, decision points.
**Vision concern:** May be reading on phone, tablet, or printed page. Layout must be responsive.

**The Older Professional**

**Context:** 20+ years in infosec, may have age-related vision changes (presbyopia, reduced contrast sensitivity).
**Needs:** Everything both audiences need, plus accommodation for reduced visual acuity.
**Vision concern:** This is the primary reason the font-size floor exists. 10px text that “looks fine” to a 25-year-old developer may be unreadable to the 55-year-old CISO.

## 3. Enforcement

### Automated Checks

| CHECK | TOOL | THRESHOLD |
|---|---|---|
| Font size floor | CSS audit | Zero sub-0.75rem values |
| Contrast ratio | squirrelscan | Zero errors |

| | | |
|---|---|---|
| Text overflow | Manual viewport testing | No horizontal scrollbar at 375px, 768px, 1024px, 1440px |
| Heading hierarchy | squirrelscan | No skipped levels |
| Lighthouse accessibility | Chrome Lighthouse | Score ≥ 95 |

### Manual Review Checklist

- **CLARITY** Can a non-technical executive understand the main finding within 10 seconds?
- **CLARITY** Does every acronym have a first-use expansion or link?
- **CLARITY** Does every security finding include a “what to do” action?
- **VISION** At 50cm / 20″, can all text be read on a 13″ laptop at 100% zoom?
- **VISION** In scotopic mode, does all text pass WCAG AA contrast (4.5:1)?
- **VISION** On a 375px mobile viewport, does all content fit without horizontal scroll?
- **VISION** Are interactive elements large enough for touch targets (44×44px)?

## 4. CSS Reference Values

Enforced values as of v26.38.26:

| PROPERTY | VALUE | CONTRAST RATIO | STANDARD |
|---|---|---|---|
| Font size floor | `0.75rem` (12px) | — | All `.u-fs-*` utility classes enforce this minimum |
| `.text-muted` | `#9ca3af` on `#0d1117` | ~7.45:1 | WCAG AA (requires 4.5:1) |
| `.card.bg-dark .text-muted` | `#adb5bd` on `#21262d` | ~8.8:1 | WCAG AA |
| Secondary text minimum | `#8b949e` on `#0d1117` | ~6.15:1 | Replaces prior `#6c757d` (~4.04:1) |

## 5. Connection to Project Philosophy

**[ACCOUNTABILITY]** — If the text is unreadable, the finding is unaccountable: it exists but cannot be acted upon.

**[ZERO_WASTE]** — Text that cannot be read is wasted: it consumes space without delivering value.

**[VERBOSE+++++]** — Clarity does not mean brevity. It means every word earns its place by advancing the reader's understanding.

**Vision as Ethics** — Causing eye strain to a reader who came for security intelligence is a design failure with real consequences.

---

**OWL SEMAPHORE SYSTEM — CLASSIFICATION LEDGER**

**NORMATIVE**

$T = I \quad \det = +1$

$(x, y) \to (x, y)$

"This is the standard."

**RFC 2119 MUST / SHALL**

**NON-NORMATIVE**

$T = \sigma_V \quad \det = -1$

$(x, y) \to (-x, y)$

"This reflects the standard."

**Informative / Advisory (NOTE)**

**CRITICAL**

$T = C_2 \quad \det = +1$

$(x, y) \to (-x, -y)$

"This inverts the standard."

**RFC 2119 MUST NOT / SHALL NOT**

**COMPOSITION IDENTITIES IN O(2)**

$\sigma \circ \sigma = I \quad \cdot \quad R(\theta_1) \circ R(\theta_2) = R(\theta_1+\theta_2) \quad \cdot \quad \sigma \circ R(\theta) = \sigma_\theta$

---

"A score without a citation is an opinion. A finding without readability is invisible."

© 2024–2026 IT Help San Diego Inc. — DNS Security Intelligence

**Related documents:** Founder's Manifesto · Philosophical Foundations · Methodology Paper · Reference Library · Sources & Citations